

Amjad Haider

Software Engineer

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Kurt-Schumacher-Straße 12, 67663 Kaiserslautern, Germany · Willing to relocate within Germany



PROFILE

Software engineer with over 4 years of experience building software for robots, vehicles and embedded systems. Strong background in sensor data processing and machine learning, in particular model compression and reinforcement learning on edge hardware, published at two robotics conferences. Confident across embedded systems, backend services and web development, with hands-on experience in IoT protocols, containerisation and applied AI tooling. Automotive experience from two internships at Volkswagen. Willing to relocate within Germany.

SKILLS

Robotics, Perception	ROS 2, Micro-ROS, MoveIt 2, SLAM, sensor fusion, Dora-RS, Hephaestus, Gymnasium, OpenCV, MuJoCo
AI	PyTorch, TensorFlow, scikit-learn, reinforcement learning, model predictive control, pruning and quantisation, knowledge distillation, ONNX, Weights & Biases, PyOD, Matplotlib, Pandas, NumPy
LLM, Agentic AI	LangChain, Azure AI, Dify, Langfuse, Ollama, Groq, Nomic, Streamlit, Gradio, MCP, Retrieval-Augmented Generation (RAG)
Embedded, IoT	MQTT, Thingsboard, Grafana, TCP/IP, REST, CRC-16/MCRF4xx, NVIDIA Jetson and Nano, Raspberry Pi, ESP32, FastAPI
Programming	C++ (17, STL), Python, Java, C#, JavaScript, MATLAB, SQL, Rust
Messaging, Streaming	Kafka, MQTT
Web, Backend, Frontend	React, Angular, Node.js, Vite, Playwright, Qt, ImGui, FastAPI
Testing, Performance	JUnit, Mockito, GTest, Valgrind, perf, pytest
DevOps, Cloud, CI/CD	Linux, Git, CMake, Docker, Kubernetes, AWS, Azure, Jenkins, GitHub Actions, Confluence
Automation	Python scripting, Bash, n8n, cron jobs, systemd services, Ansible
Simulation	MATLAB, Simulink, Gazebo, MuJoCo, Unity, Unreal Engine, Godot, Blender
Research Tools	LaTeX, Zotero, Mendeley, Overleaf, Citavi
Certification	Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103)
Languages	German B2 (completed, C1 course booked) · English C1 · Driving licence, category B

PROFESSIONAL EXPERIENCE

Freelance Embedded Software Engineer

Feb 2026 - Jul 2026

iDTRONIC GmbH, Ludwigshafen

- Built a TCP proxy server in Node.js for protocol parsing of GPS tracker ECUs.
- Implemented CRC-validated communication parsers (CRC-16/MCRF4xx and XOR) for embedded systems.
- Delivered a real-time telemetry pipeline on the Thingsboard IoT platform with an MQTT uplink decoder.
- Managed the SQL database for storage, querying and analysis of telemetry data.

Master's Thesis Student, Control Engineering

Mar 2025 - Sep 2025

RPTU Kaiserslautern-Landau, Chair of Electromobility

- Wrote the thesis "Resource-Aware Pruning in Temporal Difference Model Predictive Control: Improving Efficiency while Preserving Stability", graded 1.0.
- Analysed compression techniques (pruning, quantisation, knowledge distillation, low-rank factorisation) for edge devices (NVIDIA Jetson, NVIDIA Nano, Raspberry Pi).
- Developed a component-aware pruning approach in PyTorch.
- Compressed the model by 20.1 percent with minimal performance loss (reward 849.3 against 858.6).

Research Assistant

May 2024 - Nov 2025

RPTU Kaiserslautern-Landau, simulation and control software for robotic systems

- Developed and tested sensor data simulation platforms using ROS.
- Implemented MATLAB simulations and control for ARM manipulators with Raspberry Pi.
- Trained car control policies in CarRacing-v3 with offline RL using CleanRL and Stable Baselines.
- Implemented control software in Python, C++ and Java for real and simulated robotics applications.
- Set up IoT networks with ESP32 and Raspberry Pi, streaming data over MQTT and evaluating it on a platform.
- Built a React app and operated it alongside other services via Docker.
- Designed IoT architectures with MQTT for simulation using Python and Docker.

Intern, Navigation Systems

May 2024 - Aug 2024

Volkswagen AG, Wolfsburg

- Analysed large data volumes and data traces using systemd services and tmux.
- Evaluated competing navigation approaches in ICAS3 systems with PyTorch.
- Developed desktop application interfaces with the Qt framework.
- Developed an Ethernet interface between the navigation system (debug) and a Linux system.
- Decoded and prepared Ethernet traces and forwarded them to downstream processes and the cloud.

Intern, Autonomous Driving Functions

Nov 2023 - Apr 2024

Volkswagen Commercial Vehicles, Wolfsburg

- Improved safety and efficiency of autonomous driving using Amazon AWS.
- Developed Operational Design Domain (ODD) analysis strategies and documented them in Confluence.
- Implemented a Confluence dashboard presenting ODD-dependent technical specifications.
- Researched ISO 34503:2023 on the safety of autonomous driving functions.

Research Assistant

Feb 2023 - Oct 2023

RPTU Kaiserslautern-Landau, autonomous vehicles with sensor fusion

- Contributed to a knowledge graph project using SysML and LLMs, implemented in Python (Django) and JavaScript.
- Built a database driven by small language models (SLMs) for intelligent product development.
- Developed the autonomous unmanned surface vehicle "aStrider" for environmental monitoring and sensor-fusion navigation.
- Co-developed the autonomous minibus "AutoBus" for passenger transport in pedestrian zones and on campus.
- Supported student teams technically in lab exercises, configuring and programming model drones.
- Integrated sensor interfaces in C++ with ROS for autonomous robotic systems.

Software Engineer

Nov 2018 - Aug 2021

Xitadel, Bengaluru, India, automation for FEA and mesh generation

- Automated finite element analyses with Python scripts and built dashboards with Python and Qt.
- Improved the mid-mesh generation process using ANSA.
- Implemented and debugged complex software modules.

EDUCATION

Master of Science, Commercial Vehicle Technology, grade 1.9

Oct 2021 - Nov 2025

[RPTU Kaiserslautern-Landau, Germany](#)

Focus on software development, AI and autonomous systems. Courses: Autonomous Systems, Autonomous Mobile Robots (AMR), Electrics and Information Technology in Commercial Vehicles, Vehicle Vibrations, Vehicle Transmissions, Automotive Production, Electric and Hybrid Vehicles. Class representative, 2021 cohort.

Bachelor of Technology, Mechanical Engineering, grade 2.0 (German scale)

Sep 2013 - Apr 2017

[Cochin University of Science and Technology \(CUSAT\), Kerala, India](#)

Courses: Mechatronics, Electrical Engineering, Robotics and Control Engineering, Machine Design, CAD, Automotive Engineering, Vibration and Noise Control, Operations Management. Treasurer of the student association "FORUM".

Job-related German course, DaF level B2

Feb 2026 - Jul 2026

[Volkshochschule Kaiserslautern, Germany](#)

PUBLICATIONS

- Sundaram, G., Ulmen, J., **Haider, A.** and Görges, D. (2025). *COMponent-Aware Pruning for Accelerated Control Tasks in Latent Space Models*. IEEE/SICE International Symposium on System Integration (SII 2026).
- Sundaram, G., Ulmen, J., **Haider, A.** and Görges, D. (2025). *Application-Specific Component-Aware Structured Pruning of Deep Neural Networks via Soft Coefficient Optimization*. 22nd International Conference on Advanced Robotics (ICAR 2025).
- Keen, H. E., **Haider, A.** and Berns, K. (2023). *Denoising and Segmentation of SONAR Images for Rescue Operations*. Proceedings of the 56th International Symposium on Robotics (ISR Europe 2023).